Axicabtagene ciloleucel (Yescarta®)

Axicabtagene ciloleucel (Yescarta®) is a cancer drug made from your own white blood cells. It is used to treat lymphoma.

On this page

- What is axicabtagene ciloleucel?
- Preparing your treatment
- Having axicabtagene
- About side effects
- Side effects while treatment is being given
- Very common side effects
- Other side effects
- Other important information
- How we can help

What is axicabtagene ciloleucel?

Axicabtagene ciloleucel is a cancer drug made from your own white blood cells. It is used to treat a type of blood cancer called lymphoma. It may sometimes be used to treat other cancers.

This treatment is also called Yescarta® or Axicel. In this information we call it axicabtagene.

Axicabtagene is a type of treatment called CAR T-cell therapy. It is made from white blood cells called T-cells. T-cells are an important part of the body's immune system. They help protect us from infection or abnormal cells such as cancer cells.

Your cancer team will talk to you about this treatment and its possible side effects before you agree (consent) to have treatment. They will explain when your axicabtagene will be ready.

It is best to read this information with our general information about CAR T-cell therapy and the type of cancer you have.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the electronic Medicines Compendium (eMC) website, which has patient information leaflets (PIL) for individual drugs.

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Preparing your treatment

CAR T-cell therapy takes several weeks to make and involves the following steps:

- White blood cells are collected from your blood. This process is called leukapheresis.
- The cells are sent to a laboratory that makes CAR T-cell therapy.
- The cells are changed to help them recognise and attack cancer cells.
- More of the changed cells are grown to make your treatment.
- Your changed cells (called CAR T-cells) are sent back to the hospital where you will have the treatment.

CAR T-cell therapies are carefully made under strict conditions. But sometimes the process does not succeed in making a treatment. If this happens, your cancer team will talk to you about your options.

While the CAR T-cell therapy is being made, some people will need additional treatments to keep the cancer cells under control. This is called bridging and can involve cancer drugs or radiotherapy. Your cancer team will explain if you need bridging treatments.

In the days just before you have CAR T-cell therapy, you will have chemotherapy. This reduces the white blood cells in your body so the CAR T-cell therapy can be effective. Your cancer team will explain the chemotherapy drugs you are having and give you separate information about this.

Having axicabtagene

You have a single dose of axicabtagene during a stay in hospital. You will usually stay in hospital for at least 7 days after your treatment. This means your cancer team can treat any side effects quickly. You may need to stay for longer if you do not live close to the hospital.

Before and after your treatment, your cancer team will give you information and support. Your team will include a:

- cancer doctor
- chemotherapy nurse or specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

You have axicabtagene as a drip into a vein (intravenous infusion). Your nurse will give it through one of the following:

- a cannula – a short, thin tube the nurse puts into a vein in the arm or hand
- a central line – a fine tube that goes under the skin of the chest and into a vein close by
- a PICC line – a fine tube that goes into a vein in the arm and up into a vein in the chest
- an implantable port (portacath) – a disc that goes under the skin on the chest or arm and into a vein in the chest.

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The drip takes about 10 to 20 minutes. One of the ingredients in the treatment may give off a sweetcorn smell while the treatment is being given.

Going home

Your cancer team will monitor you closely after treatment. They will tell you when it is safe for you to go home. They will ask you to stay near the hospital for the first few weeks. They will explain what this means in your area and tell you about the follow-up appointments you need.

Your cancer team may also give you medicines to take home. Take any medicines exactly as they tell you to.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Your patient alert card

They will also give you a card with information about your treatment. It is important to keep the card with you during treatment and after treatment ends. You should show it to any doctor or healthcare professional caring for you.

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Side effects while treatment is being given

Some people may have side effects while they are having this treatment or shortly after they have it.

Allergic reaction

Some people have an allergic reaction while having this treatment. Before treatment, you will have medicines to help prevent or reduce any reaction.

Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during and after your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Cytokine release syndrome (CRS)

As the CAR T-cells find and recognise the cancer cells, chemical messengers called cytokines are released. Cytokines help your immune system attack the cancer cells. But large amounts can make you feel unwell with flu-like symptoms. This is called cytokine release syndrome or CRS.

CRS usually happens within the first 10 days of treatment. But it can happen later. If you have any of the following symptoms after you go home, contact the hospital straight away on the 24-hour number:

- a high temperature
- a fast heart rate
- dizziness
- muscle or joint pain
-

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diarrhoea

- feeling sick or being sick
- difficulty breathing.

Almost everyone who has CAR T-cell therapy will get some CRS symptoms. For most people symptoms are mild. But if CRS becomes more serious, you may need treatment and support to stop it getting worse. Some people will need treatment in intensive care for a time.

Effects on the brain (ICANS)

This treatment can cause temporary effects on the brain. This is called neurotoxicity or ICANS (immune effector cell-associated neurotoxicity syndrome).

For most people the side effects are mild. For example, you may notice changes in your handwriting. Or you may feel confused. But side effects can be more serious. You may need treatment and support to stop it getting worse. Some people will need treatment in intensive care until the symptoms improve.

ICANS side effects usually happen in the first week after treatment. But they can happen after this. They may take a few weeks to improve.

You must not drive or use heavy machinery for at least 8 weeks after this treatment. If you have any of the following symptoms after you go home, contact the hospital straight away on the 24-hour number:

- difficulty speaking
- difficulty writing or doing fiddly tasks, such as fastening buttons or tying shoelaces
- confusion
- reduced consciousness (being less alert)
- tremors or shaking
- loss of balance
- headaches
- seizures (fits).

Risk of infection

This treatment affects your body's ability to fight infection. This can become a long-term effect after treatment. You are more likely to get infections. You have a higher risk of serious illness from infection.

You will have regular blood tests after treatment to check your body's ability to fight infection. Your doctor will give you antibiotics and other drugs during and after this treatment to try and reduce your risk. But it is still important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
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you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Loss of appetite

This treatment can affect your appetite. Don’t worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may

suggest changes to your diet or eating habits to help.

Constipation

This treatment can cause constipation. Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- Eat high-fibre foods, such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Feeling tired (fatigue)

This treatment can make you feel very tired and you will need a lot of rest. You may get tired easily for some months after treatment ends. Gentle exercise, like short walks, can help you feel less tired. The tiredness will slowly get better.

Difficulty sleeping (insomnia)

Sometimes this treatment can affect your sleep. Tell your doctor, nurse or pharmacist if cancer treatment makes it difficult to sleep.

Mood changes

Tell your doctor, nurse or pharmacist if you feel anxious or have mood changes you find difficult.

Skin changes

This treatment may cause a rash, which may be itchy. Always tell your doctor, nurse or pharmacist about any skin changes. They can give you advice or prescribe creams or medicines to help.

If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

Effects on the liver and kidneys

This treatment can affect how your kidneys or liver work. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your kidneys and liver are working. Contact the hospital on the 24-hour number if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Drinking fluids also helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

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Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor, nurse or pharmacist if you develop:

- a cough that does not go away
- wheezing
- breathlessness.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

Blood pressure changes

This treatment may cause high or low blood pressure. You will have regular blood pressure checks during and after treatment.

Tummy pain

You may get pain in your tummy (abdomen), or have indigestion. Your doctor, nurse or pharmacist can give you advice or treatment to help. Contact the hospital straight away on the 24-hour number if your symptoms do not go away, or get worse.

Fluid build-up

This treatment can cause a build-up of fluid in the body. This will slowly get better after treatment ends. Contact the hospital on the 24-hour number if you:

- are gaining weight
- have swelling in your face, legs or ankles.

They can give you advice and treatment to help.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Eyesight changes

This treatment may affect your eyesight or make it harder to see clearly. Contact the hospital straight away on the 24-hour number if you have eyesight changes during treatment or after it ends. Tell your doctor, nurse or pharmacist if you notice this. Do not drive or operate machinery if you have eyesight problems.

Effects on the heart

This treatment can affect how the heart works. You may have tests to check how well your heart is working. These may be done before, during and after treatment.

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Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.
- flu-like symptoms, such as body aches, pains and tiredness.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Sore mouth and throat

This treatment may cause a sore mouth and throat. You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

Tumour lysis syndrome (TLS)

Some people are at risk of developing a condition called tumour lysis syndrome (TLS). The risk is highest at the beginning of treatment. TLS happens when treatment makes large numbers of cancer cells die and break down quickly. This releases lots of waste products into the blood and can affect the kidneys and heart.

TLS can be prevented or treated. You will have regular blood tests to check for TLS. You may have:

- extra fluids through a drip
- medicines such as allopurinol tablets or rasburicase through a drip.

Drinking at least 2 litres (3½ pints) of fluid a day will also help.

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Second cancer

This treatment can increase the risk of developing a second cancer years later. This is rare. The benefits of treatment usually far outweigh this risk. Your doctor, nurse or pharmacist can give you more information.

Other important information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a blood clot. Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- herbal drugs and complementary or homeopathic therapies
- recreational drugs – for example, cannabis.

Hepatitis B reactivation

If you have had a liver infection called hepatitis B, this treatment can make it active again. Your doctor, nurse or pharmacist will talk to you about this. You may have tests for hepatitis B before, during, and after treatment.

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Driving

You must not drive or use heavy machinery for at least 8 weeks after this treatment. You may not be allowed to drive for a longer period of time if you have had neurological side effects (‘Effects on the brain’ above). Ask your cancer team for advice.

Vaccinations

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them. They may arrange for you to repeat some vaccines you have had previously. This is because CAR T-cell therapy can make previous vaccines less effective.

Cancer doctors usually recommend that people with cancer have vaccinations for flu and coronavirus (covid). They may also recommend other vaccines, such as Shingrix® for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

Contraception

Your doctor, nurse or pharmacist will advise you not to get pregnant or make someone pregnant while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to breastfeed while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can get pregnant or make someone pregnant. If you are worried about this, it is important to talk with your doctor before you start treatment.

Blood donation

You must not donate blood or other tissue after this treatment.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Show them your patient alert card. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

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If you have appointments with a dentist or optician, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Visit the electronic Medicines Compendium (eMC) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

Cancer Information Development team

Author

Date reviewed

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Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

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We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

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